

תרגיל בית 2

מבוא לתכנות מערכות 2340124

מוחמד אבו ליל 213511009

עונג וקנין 324223486

חלק יבש מטלה 2 :

:3.2

```
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW2 % g++ -DNDEBUG -std=c++17 -Wall -pedantic-errors -Werror -o dry *.cpp
zsh: no matches found: *.cpp
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW2 % cd dry
onegvaknin@h-MacBook-Pro-sl-Oneg dry % g++ -DNDEBUG -std=c++17 -Wall -pedantic-errors -Werror -o dry *.cpp
main.cpp:15:12: error: no matching constructor for initialization of 'Pirate[2]'
   15 |     Pirate strawHatCrew[2];
      |         ^
./Pirate.h:13:5: note: candidate constructor not viable: requires single argument 'name', but no arguments were provided
   13 |     Pirate(const string& name);
      |     ^
./Pirate.h:8:7: note: candidate constructor (the implicit copy constructor) not viable: requires 1 argument, but 0 were provided
    8 | class Pirate {
      |       ^
1 error generated.
onegvaknin@h-MacBook-Pro-sl-Oneg dry %
```

השגיאה התרחשה מכיוון שב main.cpp ניסו ליצור מערך אובייקטים מסוג Pirate אבל למחלקה Pirate לא היה בנאי דיפולטיבי. יש צורך בבנאי דיפולטיבי מכיוון שכל איברי המערך מאותחלים באמצעותו. כדי לפתור את השגיאה הוספנו לבנאי אתחול לשדה השם (name = "") בבנאי המחלקה Pirate. במימוש של הבנאי אתחלנו את name ל string ריק, כך שאם הבנאי לא מקבל כלום הוא יוצר עצם (אובייקט) עם name ריק באופן הבא:

```
public:
    Pirate(const string& name="");
```

ניתן לראות שזה אכן פתר את הבעיה ועבר קומפילציה:

```
onegvaknin@h-MacBook-Pro-sl-Oneg dry % git add .
onegvaknin@h-MacBook-Pro-sl-Oneg dry % git commit -m "Compilation"
[master 16c2a8b] Compilation
 8 files changed, 680 insertions(+)
 create mode 100644 dry/.idea/.gitignore
 create mode 100644 dry/.idea/dry.iml
 create mode 100644 dry/.idea/editor.xml
 create mode 100644 dry/.idea/modules.xml
 create mode 100644 dry/.idea/vcs.xml
 create mode 100644 dry/Pirate.cpp
 create mode 100644 dry/Pirate.h
 create mode 100644 dry/main.cpp
onegvaknin@h-MacBook-Pro-sl-Oneg dry % g++ -DNDEBUG -std=c++17 -Wall -pedantic-errors -Werror -o dry *.cpp
onegvaknin@h-MacBook-Pro-sl-Oneg dry % XX
```

3.4: בסעיף זה ניסינו למשוך את הבראנץ devil fruit ולמזג את השינויים שעליו עם השינויים

שאנחנו עשינו.

התהליך:

```
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git remote add devil_fruit https://github.com/CS234124/HW2_2425B.git
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git remote -v
devil_fruit      https://github.com/CS234124/HW2_2425B.git (fetch)
devil_fruit      https://github.com/CS234124/HW2_2425B.git (push)
origin           https://github.com/MohamedAbolell/Matam_HW_2.git (fetch)
origin           https://github.com/MohamedAbolell/Matam_HW_2.git (push)
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git fetch devil_fruit devil_fruit
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (17/17), done.
remote: Total 18 (delta 3), reused 16 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (18/18), 3.36 KiB | 312.00 KiB/s, done.
From https://github.com/CS234124/HW2_2425B
 * branch                devil_fruit -> FETCH_HEAD
 * [new branch]          devil_fruit -> devil_fruit/devil_fruit
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git branch -r
  devil_fruit/devil_fruit
  origin/HEAD -> origin/master
[ origin/master
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 %
```

```
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git checkout -b devil_fruit devil_fruit/devil_fruit
branch 'devil_fruit' set up to track 'devil_fruit/devil_fruit'.
Switched to a new branch 'devil_fruit'
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git branch
* devil_fruit
[ master
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git push origin devil_fruit
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 8 threads
Compressing objects: 100% (18/18), done.
Writing objects: 100% (18/18), 3.38 KiB | 3.38 MiB/s, done.
Total 18 (delta 3), reused 5 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), done.
remote:
remote: Create a pull request for 'devil_fruit' on GitHub by visiting:
remote:   https://github.com/MohamedAbolell/Matam_HW_2/pull/new/devil_fruit
remote:
To https://github.com/MohamedAbolell/Matam_HW_2.git
 * [new branch]          devil_fruit -> devil_fruit
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 %
```

כאן מוצג הקונפליקטים:

```
* master
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git pull origin master
From https://github.com/MohamedAbolell/Matam_HW_2
 * branch      master      -> FETCH_HEAD
Already up to date.
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git merge devil_fruit
fatal: refusing to merge unrelated histories
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git merge devil_fruit --allow-unrelated-histories
Auto-merging dry/Pirate.cpp
CONFLICT (add/add): Merge conflict in dry/Pirate.cpp
Auto-merging dry/Pirate.h
CONFLICT (add/add): Merge conflict in dry/Pirate.h
Auto-merging dry/main.cpp
CONFLICT (add/add): Merge conflict in dry/main.cpp
CONFLICT (add/add): Merge conflict in wet/Utilities.cpp
CONFLICT (add/add): Merge conflict in wet/Utilities.h
CONFLICT (add/add): Merge conflict in wet/test.expected
CONFLICT (add/add): Merge conflict in wet/tests.cpp
Automatic merge failed; fix conflicts and then commit the result.
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 %
```

אחרי שעשינו השינויים הדרושים ופתרנו את הקונפליקט עשינו commit חדש והעלינו לrepo :

```
Automatic merge failed; fix conflicts and then commit the result.
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git add .
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git commit -m "resolve conflivts 3.4"
[master 4823648] resolve conflivts 3.4
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 % git push origin master
Enumerating objects: 57, done.
Counting objects: 100% (57/57), done.
Delta compression using up to 8 threads
Compressing objects: 100% (33/33), done.
Writing objects: 100% (37/37), 28.26 KiB | 7.06 MiB/s, done.
Total 37 (delta 8), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (8/8), completed with 6 local objects.
To https://github.com/MohamedAbolell/Matam_HW_2.git
   82b7fbf..4823648  master -> master
onegvaknin@h-MacBook-Pro-sl-Oneg Matam_HW_2 %
```

קימפול והרצת הקוד(עם הפלט), הדפסת כל השינויים לתוך הקובץ log.txt:

```
To https://github.com/MohamedAbolell/Matam_HW_2.git
onegvaknin@h-MacBook-Pro-sl-Oneg dry % g++ -DNDEBUG -std=c++17 -Wall -pedantic-errors -Werror -o dry *.cpp
onegvaknin@h-MacBook-Pro-sl-Oneg dry % ./dry
Straw Hat crew contains:
Luffy,1000000, (Devil Fruit: Gum Gum)
Zoro,500000, (Devil Fruit: None)
onegvaknin@h-MacBook-Pro-sl-Oneg dry %
```